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Imaging findings after acute sport-related concussion in American football players: A systematic review

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ABSTRACT

Sport-related concussion (SRC) has emerged as a major public health problem. The results of brain imaging studies following SRC have raised questions about long-term neurologic health, but the clinical implications of these findings remain unknown. A systematic review of brain imaging findings after SRC was performed utilizing the following inclusion criteria: football players, brain imaging within 6 months of SRC, and sample size >5. Studies were assessed for: 1) methodology, 2) imaging outcomes, and 3) number of positive statistical comparisons. Imaging was classified as *immediate* (≤1 week post-injury) or *subacute* (>1 week to 6 months post-injury). Eleven studies met inclusion criteria. Eight of the 11 studies conducted a total of 809 comparisons of brain function, of which 149 (18%) were statistically significant. Nine of the 11 studies (82%) reported positive *immediate* findings, but were more likely to be subject to recall bias (86% vs. 0%) and to lack baseline advanced brain imaging (78% vs. 50%) than negative studies. Only 3 of 9 studies that reported *subacute* findings (33%) reported positive results, and these positive studies were also more likely to be subject to recall bias (100% vs. 40%) and to lack baseline advanced brain imaging (100% vs. 67%) than negative studies. The results of the study demonstrate the transitory nature of positive imaging findings and methodological limitations that complicate study interpretation. Further research is required to correlate imaging findings with clinical outcomes.

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1. Introduction

Sport-related concussion (SRC) is a major public health concern in the United States (U.S.). American football remains at the heart of this controversy, in part due to highly publicized cases of National Football League (NFL) players diagnosed posthumously with chronic traumatic encephalopathy (CTE) [1–5]. Football is one of the most popular sports in the U.S., with approximately 3 million youth athletes, 1.1 million high school athletes, and 100,000 college athletes playing annually [6–8]. Estimates of SRC incidence per season in American football players range from 4.0% to 9.6%, with rates varying by level of play [9]. Thus, further study and scrutiny of these injuries is needed.

According to the most recent international Concussion in Sport Group (CISG) guidelines, a SRC includes negative conventional brain imaging, including CT and T1/T2-weighted MRI [10]. Though these imaging techniques are not medically indicated on a routine basis in the event of SRC, they are often utilized for research purposes to quantify the extent of acute, subacute, and long-term injury. While SRC by definition is a transient injury, concern regarding long-term neurologic effects has been raised [11,12]. As a result, recent research has explored the use of advanced brain imaging techniques to identify subtle abnormalities after SRC that are not apparent on routine CT and MRI. These advanced imaging studies have raised important questions concerning SRC, and though many authors have reported significant findings, the clinical relevance of these findings remains unclear [13–23].

Given the gap in knowledge between advanced brain imaging findings after acute SRC and the uncertainty of their clinical significance, we sought to further evaluate these studies. The aim of the current systematic review was to summarize the advanced brain

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imaging findings after acute SRC in an exclusive cohort of American football players. A special focus was placed on assessing study quality with standardized tools for reporting risk of bias and methodologic limitations.

2. Methods

2.1. Data sources and searches

A systematic literature review (Pubmed) was performed (February 2017) to identify studies investigating acute (defined as <6 months post-injury) advanced brain imaging findings after SRC in American football players (see Fig. 1). Within the 6-month time period, studies were classified as *immediate* when performed ≤1 week post-injury or *subacute* when performed from >1 week to 6 months post-injury. The Pubmed search strategy is reported in Supplementary File 1. This reviews follows PRISMA guidelines [24].

2.2. Study selection

Inclusion criteria were (1) American football players of any level; (2) brain imaging within 6 months of SRC; and (3) sample size >5. Exclusion criteria were (1) athletes from sports other than football; (2) no brain imaging within 6 months of injury; and (3) incomplete data.

2.3. Quality assessment

A standardized risk-of-bias assessment was performed using the Newcastle-Ottawa Scale (NOS) [25]. The Cochrane Collaboration has promoted use of the NOS for critical appraisal of non-randomized case-control and cohort studies [26]. The NOS has previously been used in systematic reviews of SRC, including those that informed the 2016 Berlin Concussion in Sport Group Consensus Statement [27–30]. The NOS rates studies on the basis of selection of cases and/or controls, comparability of cases and controls, and exposure/outcome. An assessment of nine characteristics produces a score of 0 to 9. A summary qualitative rating is assigned based on quantitative score as follows: good (7–9), fair (4–6), poor (0–3). In addition to the NOS, each study was assessed on the basis of four *a priori*-determined methodologic components utilized in prior studies [28,30].

1) Control selection: For studies including controls (n = 11), we evaluated the appropriateness of the controls on the basis of four criteria: age-matched, gender-matched, participation in

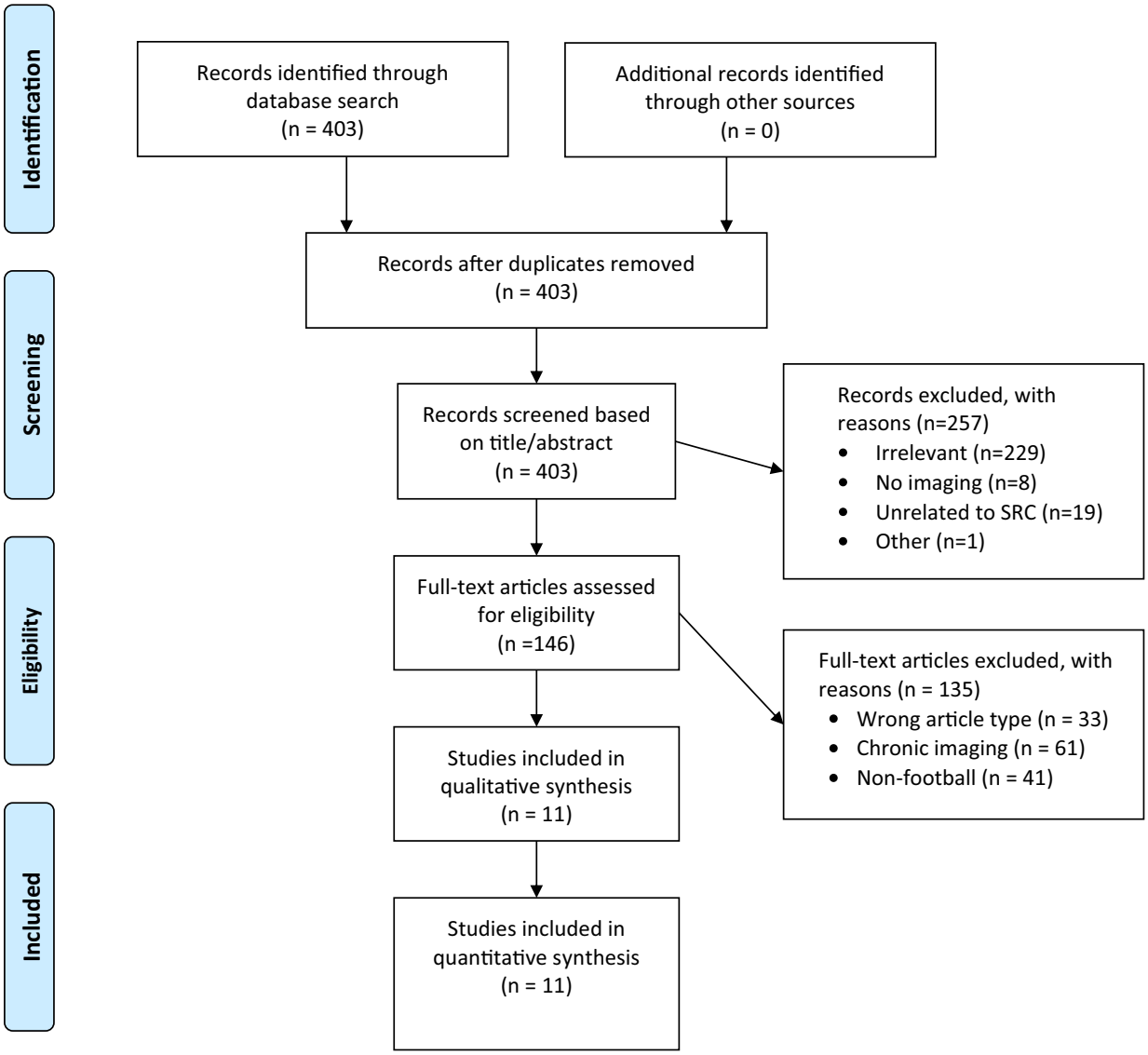

Fig. 1. Flowchart showing study selection process.

a sport with a comparable physical activity profile, and concussion history. Control groups that were matched on at least three of these four domains were considered "appropriate" or low risk for bias. Controls meeting fewer than three of these criteria were considered "inappropriate" or high risk for bias.

2) Control of Type I Error: Prior studies have identified insufficient control of Type 1 error as a potential contributor to false-positive statistical differences [31]. Only studies that reported sufficient control for multiple testing (e.g., applying corrections for family-wise error or false discovery rate) were considered "appropriate" or low risk for Type 1 error. Studies not meeting this standard were considered "inappropriate" or high risk for Type 1 error.
3) Recall bias: Prior history of concussion may be overstated or understated by athletes, either of which can lead to recall bias [32]. As such, only studies that collected prior concussion history from the medical record were considered "appropriate" or low risk for bias. Studies failing to meet this requirement (e.g. reliance solely on self-reported history) were considered "inappropriate" or high risk for bias.
4) Baseline advanced brain imaging: Studies were assessed for the acquisition of baseline advanced brain imaging before the athletes suffered a SRC. The cumulative effect of repetitive "subconcussive collisions" has been shown to alter brain imaging findings in American football players [33]. Studies with no baseline imaging were considered "inappropriate" or high risk for bias.

2.4. Data extraction

As outlined above, the methodologic integrity of each study was assessed according to the NOS, control selection, control of Type I error, recall bias, and the presence of baseline advanced brain imaging. Studies were also assessed on the basis of imaging modality and reported imaging outcomes. In an attempt to determine whether each study found a positive or negative imaging result, the number of comparisons performed in each study was tabulated. The number of statistically significant comparisons was then compared to the total number of completed comparisons. Although we did not use a strict cutoff for percentage of significant comparisons to determine whether a study reported positive findings, this information was used in our overall assessment of each study. A quantitative synthesis based on abstract and full-text review was performed by two authors (ADL and SHM). Each study was assessed for methodology and main outcome by one author (SHM) and confirmed by a senior author (SLZ).

Four distinct imaging modalities were used in the 11 studies. In order to perform a pooled analysis of individual study data, we examined the total number of imaging comparisons performed in each study. The percentage of significant comparisons was used in the determination of whether the study found a positive or negative imaging result. It should be noted that percentage of significant comparisons could not be calculated for 3 studies. The study by Jantzen et al. did not directly compare athletes who suffered a SRC to healthy controls using p-values, and the studies by Meier et al. and Wang et al. reported only brain regions with significant findings. In these cases, whether the findings constituted a positive or negative result was unknown.

2.5. Statistical analysis

Data extraction was performed for each study meeting inclusion criteria (n = 11). In addition to the methodologies of each study delineated above, total statistical comparisons were counted and percent significant comparisons were computed. Proportion tests were performed with chi-squared testing to evaluate the odds of a significant finding given methodological limitations. Significance was set at an alpha of 0.05. All statistical analyses were performed in STATA version 14 (StataCorp LP, College Station, Texas).

3. Results

3.1. Methodology

A total of 403 citations were screened and 11 studies met inclusion criteria, all of which were case-control studies. Study methodology is summarized in Table 1. Over half (54%) of the studies included only collegiate football players, while the other studies included either high school or a mix of high school and college football players. Median sample size for cases and controls was 12 (range: 3–26 and 4–27, respectively). A total of 132 American football players were compared to uninjured teammates in almost all studies (91%). Median time to initial imaging was 72 h (range: 24–168 h). All studies imaged athletes in the *immediate* timeframe, whereas 9 studies also imaged athletes in the *subacute* timeframe.

The majority of studies (54%) achieved a rating of "fair" on the NOS. Most studies demonstrated appropriate control selection (100%) and appropriate control for Type I error (73%). Conversely, a large proportion of studies displayed recall bias (67%) and lacked baseline brain imaging (73%).

3.2. Imaging modalities

A detailed summary of the imaging modalities is shown in Table 2. All studies used MRI as the imaging modality of choice, with functional MRI used in 5 of the 11 studies (45.46%). Reported outcomes included changes in blood oxygen level dependent (BOLD) activity within regions of interest (ROIs) and connectivity between ROIs. Two of the 11 studies (18.18%) used forms of diffusion-weighted imaging (diffusion tensor imaging (DTI) and/or diffusion kurtosis imaging (DKI)) to evaluate microstructural white matter changes. ^{1}H-MR spectroscopy and arterial spin labeling MRI were each utilized by two studies (18.18%) and reported changes in metabolic ratios and cerebral blood flow, respectively.

Eight of the 11 studies (72.73%) reported comparisons of brain function. These eight studies made a total of 809 comparisons, of which 149 were statistically significant (18%). Over half of the total comparisons were made by three fMRI studies that collectively made 441 comparisons, of which 67 were statistically significant (15%). The two diffusion weighted imaging studies reported 72 (out of 302) statistically significant comparisons (24%). Only one ^{1}H-MR spectroscopy and one arterial spin labeling MRI study reported comparisons (total of 9 and 12, respectively), although they both found higher percentages of significant results (33% for both). On an individual study basis, the mean number of statistically significant comparisons found was 30% (range: 3–57%).

3.3. Imaging findings

The five studies that utilized fMRI compared brain activity, as determined by signal activation by predefined ROIs or default mode network connectivity, between concussed athletes and controls [13,14,18,21,23]. Each of these studies revealed some between-group differences in brain activity. Two studies analyzed specific metabolic changes in three brain regions using MR spectroscopy, both in the immediate and subacute phases [15,17]. Both studies showed a significant difference in the concussed athletes' glucose-to-creatinine ratio (Glu:Cr) in the dorsolateral prefrontal cortex and the N-acetylaspartate-to-creatinine ratio (NAA:Cr) in the motor cortex during the acute phase, though one study showed

Table 1
Study methodologies.

Citation, year	Study Design	Level	Control	Sample size (test/control)	Time from Concussion	NOS Rating (Good 7–9), fair (4–6), or poor (1–3))	- Control selection appropriate - Control of type I error appropriate - Recall bias at play - Baseline advanced neuroimaging performed
Breedlove et al, 2012	Prospective, nested case-control	High school	Uninjured teammates	7/18	≤1 week	Good (8)	- yes - yes - yes - yes
Hammeke et al, 2013	Prospective, case-control	High school	Uninjured teammates	12/12	13 h and 7 weeks	Fair (6)	- unclear - yes - no - no
Henry et al, 2010	Retrospective, case-control	College	Uninjured teammates	12/12	1–6 days	Fair (6)	- yes - no - yes - no
Henry et al, 2011	Retrospective, case-control	College	Uninjured teammates	16/8	1–5 days and 6 months	Fair (6)	- yes - yes - unclear - no
Henry et al, 2011	Retrospective, case-control	College	Uninjured teammates	10/10	1–5 days and 6 months	Fair (6)	- yes - no - yes - no
Jantzen et al, 2004	Prospective, case-control	College	Uninjured teammates	3/4	≤1 week	Fair (6)	- yes - no - no - yes
Lancaster et al, 2016	Prospective, nested case-control	High school and college	Uninjured teammates	26/26	≤24 h and 8 days	Good (7)	- yes - yes - yes - no
Meier et al, 2015	Prospective, nested case-control	College	Uninjured teammates	17/27	~1 day, ~1 week, ~1 month	Good (7)	- yes - yes - yes - no
Talavage et al, 2014	Prospective, case-control	High school	Uninjured teammates	3/8	≤72 h	Good (8)	- yes - yes - no - yes
Wang et al, 2016	Prospective, nested case-control	High school and college	Uninjured teammates, age-matched	18/19	1 and 8 days	Good (8)	- yes - yes - yes - no
Zhu et al, 2015	Retrospective, case-control	College	Healthy, non-player active college students	8/11	24 h, ~1 week, ~1 month	Fair (6)	- unclear - yes - unclear - no

Table 2
Imaging outcomes.

Citation, year	Imaging outcome	Summary of Findings	Total Comparisons	Significant Comparisons	Percent Significant Comparisons	Immediate	Subacute
Breedlove et al, 2012	Brain activation in ROIs on fMRI	• Significant difference in brain activation in 39/116 ROIs in concussed vs. controls (mostly language processing areas)	116	39	34	Positive	N/A
Hammeke et al, 2013	Brain activation in ROIs on fMRI	• Significant group x time effect in 10 ROIs • 7/10 ROIs: underactivation 1 day and overactivation 7 weeks post-injury in concussed vs. controls • 3/10 ROIs: overactivation 1 day and underactivation 7 weeks post-injury in concussed vs. controls	60	16	27	Negative	Negative
Henry et al, 2010	Three metabolic ratios in three brain regions on MRS	• DLPFC: decreased NAA/Cr in concussed vs. controls ($p = 0.03$) • Hippocampus: no differences • M1: decreased Glu/Cr ($p = 0.004$) and NAA/Cr ($p = 0.008$) in concussed vs. controls	9	3	33	Positive	N/A
Henry et al, 2011	White matter metrics on DTI	• FA elevated in corpus callosum and right motor and sensory cortices in concussed vs. controls at both time points ($p < 0.003$) • AD elevated in R dorsal CST in concussed vs. controls at both time points ($p < 0.003$) • MD decreased in corpus callosum and ventral CST in concussed vs. controls at both time points ($p < 0.003$)	288	66	23	Positive	Positive
Henry et al, 2011	Three metabolic ratios in two brain regions on MRS	• DLPFC: main effect of group with NAA/Cr at both time points ($p = 0.017$) • M1: difference in Glu/Cr in concussed vs. controls at immediate time point ($p = 0.009$), difference in M-I/Cr in concussed vs. controls at subacute time point ($p = 0.037$)	12	5	42	Positive	Negative
Jantzen et al, 2004	Extent and amplitude of BOLD activity on fMRI	• General increase in extent and amplitude of BOLD activity in concussed vs. controls in 9 ROIs during 5 tasks • No statistical quantification of significance	45	NR	–	Negative	Negative
Lancaster et al, 2016	White matter metrics on DTI and DKI	DTI: • Widespread decreased MD in concussed vs. controls at 24 h ($p < 0.05$) that persisted at 8 days ($p < 0.05$) • Decreased AD in concussed vs. controls in numerous areas of left hemisphere at 24 h ($p < 0.05$) that became more widespread at 8 days ($p < 0.05$) • Decreased RD in concussed compared to controls in right hemisphere (CST, posterior limb of IC, SLF) at 24 h ($p < 0.05$) that became more widespread at 8 days ($p < 0.05$) • No significant difference in FA between groups at either time point DKI: • No significant difference in MK or RK between groups at either time point • Increased AK in concussed vs. controls in corpus callosum and numerous areas in right hemisphere at 24 h ($p < 0.05$) that become more widespread at 8 days ($p < 0.05$)	14	8	57	Positive	Positive
Meier et al, 2015	CBF on ASL MRI	• Lower CBF in the dmIC and STS in concussed vs. controls at T1 ($p = 0.002$ and $p = 0.046$) and T2 ($p < 0.001$ and $p = 0.04$) but not at T3 ($p = 0.68$ and $p = 0.86$)	NR	NR	–	Positive	Negative
Talavage et al, 2014	Activations on fMRI	• Significant alterations in 5/116 ROIs in concussed vs. controls on immediate scan • Significant alterations in 1/116 ROIs in concussed vs. controls on subacute scan	232	6	3	Positive	Negative
Wang et al, 2016	CBF on ASL MRI	• Lower CBF in right SMA and pre-SMA regions vs. controls at 24 h post-injury ($p < 0.05$) • Lower CBF across cortical gray matter in prefrontal, parietal, temporal, and thalamic regions vs. controls at 8 days post-injury ($p < 0.05$)	NR	NR	–	Positive	Positive
Zhu et al, 2015	DMN connectivity on rs-fMRI	• No difference in connectivity in concussed vs. control at 1 day post-injury • Decreased connectivity in bilateral frontal lobes and hippocampi in concussed vs. controls at 7 days post-injury (all $p \leq 0.004$) • No difference in connectivity in concussed vs. controls at 30 days post-injury	33	6	18	Positive	Negative

AD = axial diffusivity; ASL = arterial spin labeling; CBF = cerebral blood flow; CST = corticospinal tract; DMN = default mode network; DKI = diffusion kurtosis imaging; DTI = diffusion tensor imaging; FA = fractional anisotropy; MD = mean diffusivity; MRS = ^{1}H-MR spectroscopy; ROIs = regions of interest; rs-fMRI = resting state functional MRI; SMA = supplementary motor area.

Table 3
Comparison of study methodology by imaging outcome.

	Immediate Outcome		Subacute Outcome	
	Positive N = 9	Negative N = 2	Positive N = 3	Negative N = 6
Appropriate control selection N = 9*	8/8 (100%)	1/1 (100%)	3/3 (100%)	4/4 (100%)
Controlled for type I error N = 11	7/9 (78%)	1/2 (50%)	2/3 (67%)	4/6 (67%)
Recall bias N = 9*	1/7 (14%)	2/2 (100%)	2/2 (100%)	2/5 (40%)
Baseline advanced neuroimaging N = 11	2/9 (22%)	1/2 (50%)	0/3 (0%)	2/6 (33%)

* Excluded studies that did not provide enough information to determine classification.

no differences in the subacute phase [17]. Both studies that examined changes in cerebral blood flow (CBF) using ASL-MRI demonstrated decreased CBF in numerous brain regions in the immediate period, with return to baseline during the subacute period [20,22]. Finally, two studies used diffusion-weighted imaging (DWI) to assess changes in white matter structure after SRC. Henry et al. reported significantly elevated fractional anisotropy (FA) and axial diffusivity and significantly decreased mean diffusivity (MD) in several brain regions in concussed athletes compared to controls at both immediate and subacute time periods [16]. Lancaster et al. also reported an effect on DTI MD in concussed compared to controls at 24 h ($p < 0.05$) that persisted at 8 days ($p < 0.05$), but no significant alteration in FA [19]. Only Lancaster et al reported findings from diffusion kurtosis imaging (DKI) [19]. They found that axial kurtosis (AK) was significantly higher in the concussed group compared to controls at both the immediate and subacute time points. However, mean kurtosis (MK) and radial kurtosis (RK) did not differ significantly between the groups.

3.4. Assessment of significant findings

Table 3 provides a summary of study methodology stratified by imaging outcome. Nine of the studies (82%) reported positive immediate findings (e.g. significant differences when compared to a control group), but when compared with negative findings, were more likely to be subject to recall bias (86% vs. 0%) and to lack baseline advanced brain imaging (78% vs. 50%). Nine studies also evaluated imaging outcomes in the subacute period. Only three of these studies (33%) reported positive findings, and these positive studies were more likely to be subject to recall bias (100% vs. 40%) and to lack baseline advanced brain imaging (100% vs. 67%) than negative studies.

4. Discussion

The current systematic review evaluated 11 studies that utilized advanced brain imaging to study the effects of SRC within 6 months of injury. The studies were analyzed on the basis of methodology, imaging modality, and outcome and divided into the *immediate* and *subacute* time periods. Although most of the studies reported positive imaging findings in the *immediate* period (≤1 week post-injury), only 3 studies continued to report positive findings in the *subacute* period (>1 week to 6 months post-injury), suggesting that any neuroimaging changes are likely transient and return to baseline within months after injury in these groups of American football players. Importantly, studies reporting positive findings in either time period were more likely to be subject to recall bias and lack baseline brain imaging than negative studies.

Study methodology was examined on the basis of five criteria: risk-of-bias on the Newcastle-Ottawa Scale (NOS), control selection, control of type I error, recall bias, and presence/absence of baseline brain imaging. All studies received a score of at least "fair" on the NOS and demonstrated appropriate control selection. However, it will be important in future advanced neuroimaging research of SRC to consider control group composition, and not just the presence of any control group. For example, Wilde et al. demonstrated the importance of control group selection, revealing significant group differences (lower FA and higher MD) in the immediate period between mTBI patients and typically-developing controls, but not for mTBI patients and patients recently sustaining an orthopedic injury [34].

The 11 studies less consistently demonstrated an appropriate control of Type I error, which could have contributed to false-positive statistical differences. With conventional hypothesis testing in statistics that uses an alpha level of 0.05, we accept that the probability of Type I error (false positive) will be equal to 5%. In other words, there will be five false positives for every 100 tests performed. The necessity of correcting for multiple comparisons is essential in any research investigation and especially pertinent in advanced neuroimaging research when a large number of hypotheses involving multiple voxel-wise comparisons are studied. In one study included as part of the current review, 288 comparisons were made without correction, likely resulting in an inflation of Type I error rate [16]. As such, approximately 14 of these comparisons are expected to be statistically significant, but in fact are false positive results. Although the need for multiple comparisons is often characteristic of investigations in novel research areas, the importance of protecting against false positive findings should not be sacrificed.

In addition, almost two-thirds of the studies demonstrated recall bias and three-quarters lacked baseline advanced brain imaging. In an ideal study design of advanced brain imaging after SRC, all athletes would have their complete concussion history documented by medical professionals (and/or a review of medical records) and baseline advanced brain imaging performed before the start of the season. However, these conditions are understandably difficult to achieve. First, less than half of individuals do not seek medical treatment after SRC [35]. As a result, many high school and college athletes may have undocumented concussions that would not be found in the medical record. Second, the effect of "subconcussive" brain impacts is difficult to capture. Although preliminary studies have shown advanced brain imaging changes with repetitive subconcussive impacts, specific patterns and changes over time remain unknown [36–38]. Third, advanced brain imaging is expensive and, along with logistical concerns, may preclude the acquisition of baseline imaging prior to the start of a season. In spite of these difficulties, however, these 11 studies have made strong initial efforts to study a complex topic. That said, the limitations of these early pilot studies

with small samples, evidence of recall bias, and lack of baseline advanced brain imaging must be noted.

Regarding multiple statistical comparisons, 8 of the 11 studies reported comparisons of brain function. These studies made a total of 809 comparisons, of which 149 were statistically significant (18%). Over half of the total comparisons were made by three fMRI studies that collectively made 441 comparisons, of which 67 were statistically significant (15%) [13,14,21]. The two diffusion weighted imaging studies reported 72 statistically significant comparisons out of 302 total comparisons (24%) [16,19]. Only one ^{1}H-MR spectroscopy [15] and one arterial spin labeling MRI study [17] reported comparisons (total of 9 and 12, respectively), although they both found higher percentages of significant results (33% for both).

Overall, 9 of 11 studies (81.82%) found positive findings in the immediate period, but only 3 of 9 studies (33.33%) reported a persistent positive finding in the subacute period. While there may be transient imaging changes acutely after SRC, a number of observations weaken this conclusion. First, immediate studies with positive findings were more likely to be subject to recall bias and lack baseline imaging than studies reporting negative findings. Second, our operational definition of "positive" imaging findings may overstate the true extent of changes in brain activity in the immediate period. When interpreting positive findings, the total number of comparisons must be taken into account. Eight of 11 studies reported 149 of 809 positive comparisons (18%). Moreover, only one study reported greater than 50% positive comparisons.

The clinical significance of the positive advanced brain imaging findings remains entirely uncertain. Hammeke et al reported negative findings in the both the immediate and subacute periods and reported no significant correlation between brain activation levels and behavioral indices [14]. Henry et al reported positive findings in both the immediate and subacute periods, but also found no significant correlation between changes in diffusion tensor metrics and symptom score [16]. In a similar manner, Lancaster et al found positive imaging findings in the immediate period, yet these findings correlated with only 1 of 3 symptom severity indices [19]. Conversely, numerous studies have reported correlations between brain imaging findings and symptom severity and duration following SRC [39,40]. In order to delineate the clinical significance of advanced brain imaging findings in the immediate and/or subacute periods after SRC, further study is necessary to identify persistent changes in advanced brain imaging and correlate them not only with symptom scores (which may be not be specific to the SRC), but also with objective neurocognitive indexes to aid the clinician in appreciating possible neurobehavioral implications.

In an attempt to clarify the clinical significance of advanced brain imaging findings after SRC, Lancaster et al. recently published a longitudinal study that followed concussed athletes from the acute to chronic stage (6 months post-injury) [41]. Interestingly, they found that although diffusion metrics on DTI continued to be decreased in the concussed athletes compared to controls at 6 months post-injury, diffusion metrics on DKI had normalized. Prior studies have suggested that DKI may provide a better estimate of white matter microstructure than DTI due improved measurement through use of a more appropriate probability distribution function [42]. Taken together, these findings demonstrate the importance of critically analyzing the advanced brain imaging modality used when reporting changes in brain imaging following SRC.

There are several limitations to our study that require discussion. First, studies varied in the amount of information they provided concerning control selection. Some reported numerous variables (age, gender, activity level, concussion history), while others reported only some of these variables. We determined whether each study demonstrated appropriate control selection based on the variables provided, which could over- or underestimate how well the studies actually selected controls. Second, we were unable to assess and correct for potential confounders such as subject medical and neuropsychiatric histories and lifestyle differences (e.g. use of legal and illegal substances, mood disorders, ADHD/learning disability, migraine headaches, etc.) that may have affected associations between playing American football and brain imaging changes.

5. Conclusion

The study of acute advanced brain imaging findings after SRC is in its infancy. The heterogeneity of study design, methodological limitations, and transitory nature of positive imaging findings complicate interpretation of study results. Future neuroimaging research in SRC should look to: a) improve control selection with matching based on appropriate and relevant subject characteristics, b) adequately control for multiple comparisons, c) utilize objective outcome measures, and d) employ comparisons through control of baseline neuroimaging. Further research is required to correlate advanced imaging findings with clinical outcomes in SRC.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jocn.2018.11.022.

References

[1] McKee AC, Cantu RC, Nowinski CJ, et al. Chronic traumatic encephalopathy in athletes: progressive tauopathy after repetitive head injury. J Neuropathol Expl Neurol 2009;68(7):709–35. https://doi.org/10.1097/NEN.0b013e3181a9d503.
[2] Mez J, Daneshvar DH, Kiernan PT, et al. Clinicopathological evaluation of chronic traumatic encephalopathy in players of American football. JAMA 2017;318(4):360–70. https://doi.org/10.1001/jama.2017.8334.
[3] Omalu BI, DeKosky ST, Hamilton RL, et al. Chronic traumatic encephalopathy in a National Football League player: part II. Neurosurgery 2006;59(5):1086–92. https://doi.org/10.1227/01.NEU.0000245601.69451.27.
[4] Omalu BI, DeKosky ST, Minster RL, et al. Chronic traumatic encephalopathy in a National Football League player. Neurosurgery 2005;57(1):128–34. https://doi.org/10.1227/01.NEU.0000163407.92769.ED. discussion 128–134.
[5] Omalu BI, Hamilton RL, Kamboh MI, et al. Chronic traumatic encephalopathy (CTE) in a National Football League player: case report and emerging medicolegal practice questions. J Forensic Nurs 2010;6(1):40–6. https://doi.org/10.1111/j.1939-3938.2009.01064.x.
[6] National Council of Youth Sports. Reports on trends and participation in organized youth sports. http://www.ncys.org/pdfs/2008/2008-ncys-market-research-report.pdf. Published 2008. Accessed January 15, 2018.
[7] National Collegiate Athletic Association. Student-athlete participation 1981-1982–2012-13: NCAA sports sponsorship and participation rates report. http://www.ncaapublications.com/productdownloads/PR2014.pdf. Published 2014. Accessed January 15, 2018.
[8] National Federation of State High School Associations. Participation statistics. http://www.nfhs.org/ParticipationStatics/ParticipationStatics.aspx/. Published 2014. Accessed January 15, 2018.
[9] Dompier TP, Kerr ZY, Marshall SW, et al. Incidence of concussion during practice and games in youth, high school, and collegiate American football players. JAMA Ped 2015;169(7):659–65. https://doi.org/10.1001/jamapediatrics.2015.0210.
[10] Patricios JS, Ardern CL, Hislop MD, et al. Implementation of the 2017 Berline Concussion in Sport Group Consensus Statement in contact and collision sports: a joint position statement from 11 national and internal organisations. Br J Sports Med 2018;52(10):635–41. https://doi.org/10.1136/bjsports-2018-099079.
[11] Alosco ML, Mez J, Tripodis Y, et al. Age of first exposure to tackle football and chronic traumatic encephalopathy. Ann Neurol 2018;83(5):886–91. https://doi.org/10.1002/ana.25245.
[12] McCrea M, Guskiewicz KM, Marshall SW, et al. Acute effects and recovery time following concussion in collegiate football players: the NCAA Concussion Study. JAMA 2003;290(19):2556–63.
[13] Breedlove EL, Robinson M, Talavage TM, et al. Biomechanical correlates of symptomatic and asymptomatic neurophysiological impairment in high school football. J Biomech 2012;45(7):1265–72. https://doi.org/10.1016/j.jbiomech.2012.01.034.
[14] Hammeke TA, McCrea M, Coats SM, et al. Acute and subacute changes in neural activation during the recovery from sport-related concussion. J Int

Neuropsychol Soc 2013;19(8):863–72. https://doi.org/10.1017/S1355617713000702.
[15] Henry LC, Tremblay S, Boulanger Y, Ellemberg D, Lassonde M. Neurometabolic changes in the acute phase after sports concussions correlate with symptom severity. J Neurotrauma 2010;27(1):65–76. https://doi.org/10.1089/neu.2009.0962.
[16] Henry LC, Tremblay J, Tremblay S, et al. Acute and chronic changes in diffusivity measures after sports concussion. J Neurotrauma 2011;28(10):2049–59. https://doi.org/10.1089/neu.2011.1836.
[17] Henry LC, Tremblay S, Leclerc S, et al. Metabolic changes in concussed American football players during the acute and chronic post-injury phases. BMC Neurol 2011;11:105. https://doi.org/10.1186/1471-2377-11-105.
[18] Jantzen KJ, Anderson B, Steinberg FL, Kelso JA. A prospective functional MR imaging study of mild traumatic brain injury in college football players. AJNR Am J Neuroradiol 2004;25(5):738–45.
[19] Lancaster MA, Olson DV, McCrea MA, Nelson LD, LaRoche AA, Muftuler LT. Acute white matter changes following sport-related concussion: A serial diffusion tensor and diffusion kurtosis tensor imaging study. Hum Brain Mapp 2016;37(11):3821–34. https://doi.org/10.1002/hbm.23278.
[20] Meier TB, Bellgowan PS, Singh R, Kuplicki R, Polanski DW, Mayer AR. Recovery of cerebral blood flow following sports-related concussion. JAMA Neurol 2015;72(5):530–8. https://doi.org/10.1001/jamaneurol.2014.4778.
[21] Talavage TM, Nauman EA, Breedlove EL, et al. Functionally-detected cognitive impairment in high school football players without clinically-diagnosed concussion. J Neurotrauma 2014;31(4):327–38. https://doi.org/10.1089/neu.2010.1512.
[22] Wang Y, Nelson LD, LaRoche AA, et al. Cerebral blood flow alterations in Acute sport-related concussion. J Neurotrauma 2016;33(13):1227–36. https://doi.org/10.1089/neu.2015.4072.
[23] Zhu DC, Covassin T, Nogle S, et al. A potential biomarker in sports-related concussion: brain functional connectivity alteration of the default-mode network measured with longitudinal resting-state fMRI over thirty days. J Neurotrauma 2015;32(5):327–41. https://doi.org/10.1089/neu.2014.3413.
[24] Moher D, Liberati A, Tetzlaff J, et al. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. BMJ 2009;339:. https://doi.org/10.1136/bmj.b2535b2535.
[25] Wells G, Shea B, O'Connell D, et al. The Newcastle-Ottawa Scale (NOS) for assessing the quality of nonrandomized studies in meta-analyses. Canada: Ottawa Hospital Research Institute; 2011.
[26] Deeks J, Higgins J, Altman D, Green S. Cochrane handbook for systematic reviews of interventions version 5.1. 0 (updated March 2011). Cochrane Collab 2011.
[27] Davis GA, Anderson V, Babl FE, Gioia CC, Meehan W, Moser RS, et al. What is the difference in concussion management in children and adults? A systematic review. Br J Sports Med 2017;51(12):949–57. https://doi.org/10.1136/bjsports-2016-097415.
[28] Feddermann-Demont N, Echemendia RJ, Schneider KJ, Solomon GS, Hayden KA, Turner M, et al. What domains of clinical function should be assessed after sport-related concussion? A systematic review. Br J Sports Med 2017;51(11):903–18. https://doi.org/10.1136/bjsports-2016-097403.
[29] Kamins J, Bigler E, Covassin T, Henry L, Kemp S, Leddy JJ, et al. What is the physiological time to recovery after concussion? A systematic review. Br J Sports Med 2017;51(12):935–40. https://doi.org/10.1136/bjsports-2016-097464.
[30] Tarnutzer AA, Straumann D, Brugger P, Feddermann-Demont N. Persistent effects of playing football and associated (subconcussive) head trauma on brain structure and function: a systematic review of the literature. Br J Sports Med 2017;51(22):1592–604. https://doi.org/10.1136/bjsports-2016-096593.
[31] Rutherford A, Stephens R, Potter D. The neuropsychology of heading and head trauma in Association Football (soccer): a review. Neuropsychol Rev 2003;13(3):153–79.
[32] Kerr ZY, Mihalik JP, Guskiewicz KM, Rosamond WD, Evenson KR, Marshall SW. Agreement between athlete-recalled and clinically documented concussion histories in former collegiate athletes. Am J Sports Med 2015;43:606–13. https://doi.org/10.1177/0363546514562180.
[33] Slobounov SM, Walter A, Breiter HC, Zhu DC, Bai X, Bream T, et al. Talavage Tm. The effect of repetitive subconcussive collisions on brain integrity in collegiate football players over a single season: A multi-modal neuroimaging study. Neuroimage Clin 2017;14:708–18. https://doi.org/10.1016/j.nicl.2017.03.006.
[34] Wilde EA, Ware AL, Li X, Wu TC, McCauley SR, Barnes A, et al. Orthopedic injured versus uninjured comparison groups for neuroimaging research in mild traumatic brain injury. J Neurotrauma 2018. https://doi.org/10.1089/neu.2017.5513.
[35] Setnik L, Bazarian JJ. The characteristics of patients who do not seek medical treatment for traumatic brain injury. Brain Inj 2007;21(1):1–9.
[36] Bailes JE, Petraglia AL, Omalu BI, Nauman E, Talavage T. Role of subconcussion in repetitive mild traumatic brain injury: a review. J Neurosurg 2013;119(5):1235–45. https://doi.org/10.3171/2013.7.JNS121822.
[37] Bazarian JJ, Zhu T, Blyth B, Borrino A, Zhong J. Subject specific changes in brain white matter on diffusion tensor imaging after sports-related concussion. Magn Reson Imaging 2012;30:171–80. https://doi.org/10.1016/j.mri.2011.10.001.
[38] Talavage TM, Nauman E, Breedlove EL, Yoruk U, Dye AE, Morigaki K, et al. Functionally-detected cognitive impairment in high school football players without clinically-diagnosed concussion. J Neurotrauma 2014;31(4):327–38. https://doi.org/10.1089/neu.2010.1512.
[39] Chen SHA, Kareken DA, Fastenau PS, Trexler LE, Hutchins GD. A study of persistent post-concussion symptoms in mild head trauma using positron emission tomography. J Neurol Neurosurg Psych 2003;74:326–32.
[40] Pardini JE, Pardini DA, Becker JT, Dunfee KL, Eddy WF, Lovell MR, et al. Postconcussive symptoms are associated with compensatory recruitment during a working memory task. Neurosurgery 2010;67:1020–7. https://doi.org/10.1227/NEU.0b013e3181ee33e2. discussion 1027–8.
[41] Lancaster MA, Meier TB, Olson DV, McCrea MA, Nelson LD, Muftuler LT. Chronic differences in white matter integrityy following sport-related concussion as measured by diffusion MRI: 6-month follow-up. Hum Brain Mapp 2018. https://doi.org/10.1002/hbm.24245.
[42] Lanzafame S, Giannelli M, Garaci F, Floris R, Duggento A, Guerrisi M, et al. Differences in Gaussian diffusion tneosr imaging and non-Gaussian diffusion kurtosis imaging model-based estimates of diffusion tensor invariants in the human brain. Med Phys 2016;43(5):2464. https://doi.org/10.1118/1.4946819.